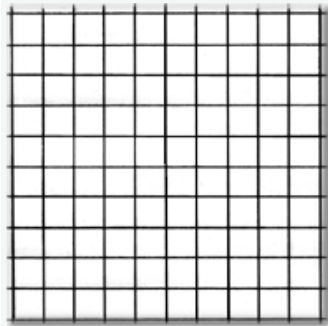
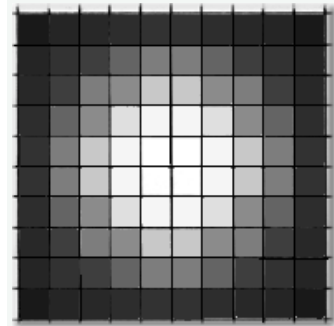


CCD - Charged Coupled Device

- Primary detector used in astronomy

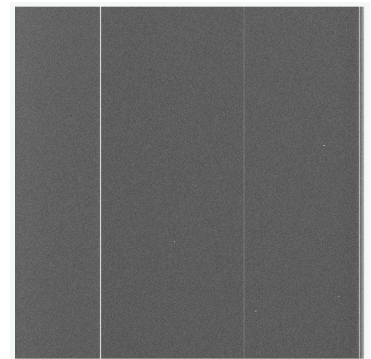


25	37	40	45	50	50	45	40	37	25
37	50	62	100	125	125	100	62	50	37
40	62	125	139	200	200	139	125	62	40
45	100	139	223	245	245	223	139	100	45
50	125	200	245	255	255	245	200	125	50
50	125	200	245	255	255	245	200	125	50
45	100	139	223	245	245	223	139	100	45
40	62	125	139	200	200	139	125	62	40
37	50	62	100	125	125	100	62	50	37
25	37	40	45	50	50	45	40	37	25



Bias Frame

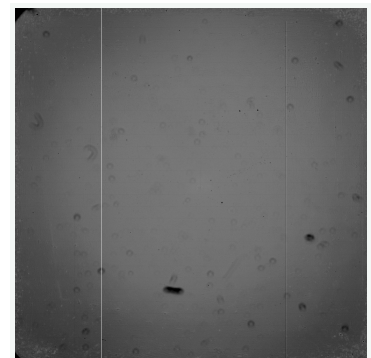
- Exposure of 0 seconds
- An image of the detector noise
- Noise is consistent with every exposure of the camera
- Take multiple frames and average them together to get “real” noise



Bias Frame

Flat Frame

- Image of bright, smooth (“flat”) field
- Helps remove artifacts associated with equipment and environment (e.g. dust)
- Flat frames need to be taken for each filter used



Flat Frame

Cosmic Rays

- High energy particles that hit the detector
- Look like very bright spots in the image

Data reduction process

- Create an average of the bias frames
- Subtract average bias from all flat frames and science frames
- Mask cosmic rays and bad pixels

$$\text{Final Image} = \frac{\text{Data} - \text{Bias}}{\text{Flat} - \text{Bias}}$$